

7. (a) Describe the main features of the spectrum of a star and state where in the star they arise. [2]

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- (b) The table gives some information about two stars.

| Star   | Luminosity / W       | Distance from Earth / m |
|--------|----------------------|-------------------------|
| Sirius | $9.7 \times 10^{27}$ | $8.1 \times 10^{16}$    |
| Vega   | $1.5 \times 10^{28}$ | $2.4 \times 10^{17}$    |

- (i) Determine the ratio:

$$\frac{\text{Intensity of radiation reaching Earth from Sirius}}{\text{Intensity of radiation reaching Earth from Vega}}$$

[3]

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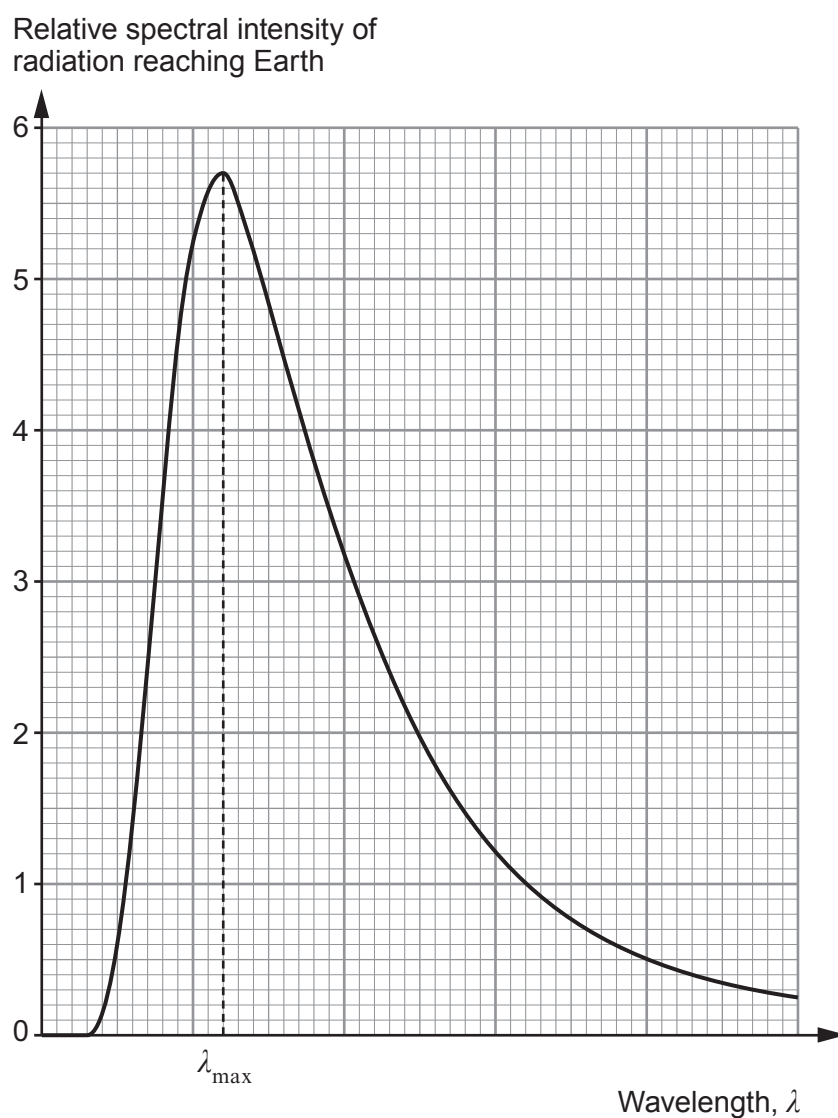
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- (ii) The two stars have similar surface temperatures. Below is a black body radiation curve for Sirius. Using your answer to (b)(i), sketch the expected black body curve for Vega on the same axes. [2]



- (iii) Determine  $\lambda_{\text{max}}$ , given that the surface area of Sirius is  $1.8 \times 10^{19} \text{ m}^2$ . [4]

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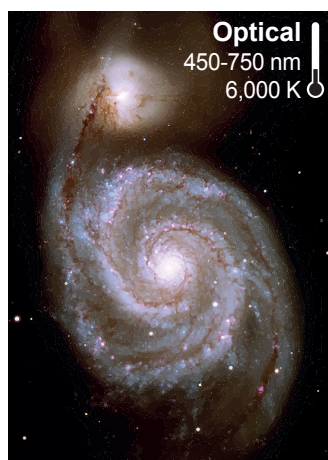
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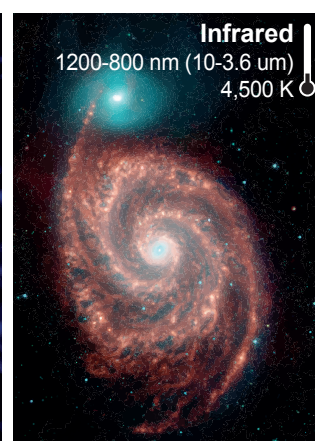
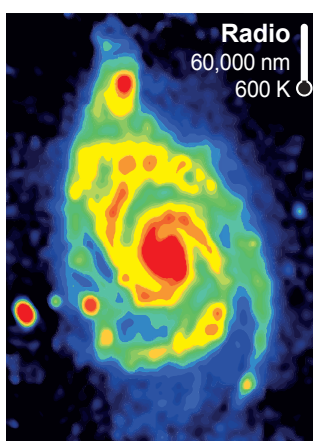
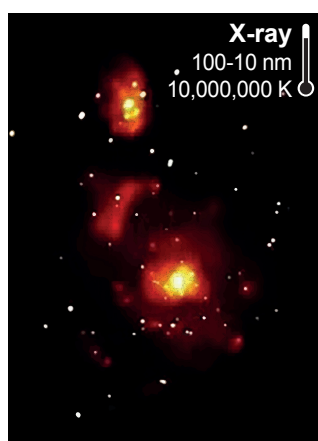
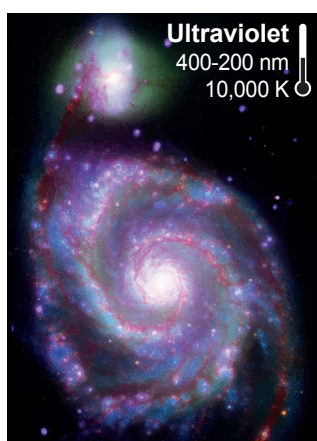
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- (c) The image below is of the whirlpool galaxy, M51 (or NGC 5194). This is one of the first galaxies to be photographed by astronomers.



Subsequent images of the **same galaxy** are shown below.



Describe how these developments in observational astronomy have advanced the study of the whirlpool galaxy. [3]

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END OF PAPER

